relativity, whose effects become significant under extremes of speed or gravity, the new physics of quantum mechanics reveals itself abundantly only in another extreme situation: the realm of the extremely tiny. But there is a sharp distinction between the upheavals of relativity and those of quantum mechanics. The weirdness of relativity arises because our personal experience of space and time differs from the experience of others. It is a weirdness born of comparison. We are forced to concede that our view of reality is but one among many—an infinite number, in fact—which all fit together within the seamless whole of spacetime.

Quantum mechanics is different. Its weirdness is evident without comparison. It is harder to train your mind to have quantum mechanical intuition, because quantum mechanics shatters our own personal, individual conception of reality.

The World According to the Quantum

Every age develops its stories or metaphors for how the universe was conceived and structured. According to an ancient Indian creation myth, the universe was created when the gods dismembered the primordial giant Purusa, whose head became the sky, whose feet became the earth, and whose breath became the wind. To Aristotle, the universe was a collection of fifty-five concentric crystalline spheres, the outermost being heaven, surrounding those of the planets, earth and its elements, and finally the seven circles of hell.[1] With Newton and his precise, deterministic mathematical formulation of motion, the description changed again. The universe was likened to an enormous, grand clockwork: after being wound and set into its initial state, the clockwork universe ticks from one moment to the next with complete regularity and predictability.

Special and general relativity pointed out important subtleties of the clockwork metaphor: there is no single, preferred, universal clock; there is no consensus on what constitutes a moment, what constitutes a *now.* Even so, you can still tell a clockworklike story about the evolving universe. The clock is your clock. The story is your story. But the universe unfolds with the same regularity and predictability as in the Newtonian framework. If by some means you know the state of the